



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

PREFLASH COLUMN STEAM HEATER CONDENSATE PUMPS (P-2308A/B)

Docu. No. : 14780-8120-PS-P2308
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
Page : 1 / 7

TITLE :

PREFLASH COLUMN STEAM HEATER CONDENSATE PUMPS

(P-2308A/B)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-P2308
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 23, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A2	FEB 25, 2014	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 11, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUL 30, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

This Document is the property of POSCO Engineering Co., Ltd. Therefore, it shall not be released to Any third party without permission of an authorized personnel of the POSCO Engineering Co., Ltd.

[illegible]

UOP LLC 25 East Algonquin Road Des Plaines, Illinois 60017-5017 USA SEALLESS PUMPS AND DRIVERS		PROJECT SPECIFICATION							
		963766 - 506				SHEET 3 / 7			
		REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL						
1	31-May-12	MW	MEH						
2	14-Aug-12	MW	MEH						
3	16-Oct-13	MW	DRL						

APPLICABLE TO: ☐ PROPOSALS ☐ PURCHASE ☒ AS BUILT			
FOR _____	UNIT _____		
SITE _____	SERVICE Preflash Column Steam Heater Condensate Pumps		
NO. REQ <u>2</u> PUMP SIZE _____	TYPE Sealless Mag. Drive		NO. STAGES <u>1</u>
MANUFACTURER _____	MODEL _____		SERIAL NO. _____

DIMENSIONS PER: ☐ SI UNITS ☐ US UNITS EQUIPMENT TO COMPLY WITH APPLICABLE: ☐ ISO ☐ US STANDARDS			
NOTES: INFORMATION BELOW TO BE COMPLETED: ☐ BY PURCHASER ☐ BY MANUFACTURER ☒ BY MANUFACTURER OR PURCHASER			
☐ GENERAL (5.15.1.1)			
PUMPS TO OPERATE IN (PARALLEL)	NO. MOTOR DRIVEN <u>2</u>	NO. TURBINE DRIVEN _____	
(SERIES) WITH _____	PUMP ITEM NO. P-2308A/B	PUMP ITEM NO. _____	
GEAR ITEM NO. _____	MOTOR ITEM NO. _____	TURBINE ITEM NO. _____	
GEAR PROVIDED BY _____	MOTOR PROVIDED BY _____	TURBINE PROVIDED BY _____	
GEAR MOUNTED BY _____	MOTOR MOUNTED BY _____	TURBINE MOUNTED BY _____	
GEAR DATA SHT. NO. _____	MOTOR DATA SHT. NO. _____	TURBINE DATA SHT. NO. _____	

☐ OPERATING CONDITIONS (6.1.3)				☐ LIQUID			
CAPACITY, NORMAL <u>16.8</u> m³/h	RATED <u>18.0</u> m³/h			TYPE OR NAME OF LIQUID <u>Condensate</u>			
OTHER <u>7.45</u> m³/h				PUMPING TEMP °C NORMAL <u>117</u> MAX. _____ MIN. _____			
SUCTION PRESS MAX./RATED <u>7.424</u> / <u>1.235</u> kg/cm²[g]				VAPOR PRESSURE <u>1.88</u> kg/cm²[a] @ <u>117</u> °C			
DISCHARGE PRESSURE <u>2.764</u> kg/cm²[g]				TEMPERATURE-VAPOR PRESSURE CURVE NO: _____			
DIFFERENTIAL PRESSURE <u>1.53</u> kg/cm²				RELATIVE DENSITY (SPECIFIC GRAVITY):			
DIFF. HEAD <u>16.2</u> m	NPSHA <u>4.1</u> m			NORMAL <u>0.947</u> MAX _____ MIN _____			
PROCESS VARIATIONS (7.1.2) _____				SPECIFIC HEAT, Cp _____ 1.01 kcal/(kg.°C)			
STARTING CONDITIONS (7.1.3) _____				VISCOSITY <u>0.239</u> cP @ <u>117</u> °C			
SERVICE: ☒ CONT. ☐ INTERMITTENT (STARTS/DAY) _____				MAX. VISCOSITY _____ (cP) @ _____ °C			
☐ PARALLEL OPERATION REQ'D (6.1.12)				TEMPERATURE-VISCOSITY CURVE NO: _____			
☐ SITE DATA				THERMAL EXPANSION: _____ mm/m/°C			
LOCATION: (6.1.29)				THERMAL CONDUCTIVITY: <u>0.586</u> kcal/(h.m.°C)			
☐ INDOOR ☐ HEATED ☐ UNDER ROOF				CORROSIVE/EROSIVE AGENT _____ (6.11.7)			
☒ OUTDOOR ☒ UNHEATED ☐ PARTIAL SIDES				CORROSION ALLOWANCE (6.3.3) _____ mm			
☒ GRADE ☐ MEZZANINE ☐ _____				☐ CHLORIDE PRESENT: CONCENTRATION (PPM) _____ (7.3.2.6)			
☐ ELECTRIC AREA CLASSIFICATION (6.1.24)				☐ WET H2S PRESENT: CONCENTRATION (PPM) _____ (6.11.11)			
CL <u>1</u> GR <u>IIA</u> DIV <u>ZONE 2</u>				☐ SOLIDS PRESENT (6.2.1): CONCENTRATION (% by wt.) _____			
☐ WINTERIZATION REQ'D ☐ TROPICALIZATION REQ'D. (7.2.4.5)				PARTICLE SIZE _____ HARDNESS _____			
SITE DATA (6.1.29) SEE GENERAL NOTES ON SHEET 6 *				SIZE DISTRIBUTION _____			
☐ ALTITUDE <u>0</u> BAROMETER _____				LIQUID (6.1.3) ☐ HAZARDOUS ☐ FLAMMABLE			
☐ RANGE OF AMBIENT TEMPS: MIN/MAX. _____ / _____ °C				POLYMERIZATION CHARACTERISTICS (6.2.1): _____			
☐ RELATIVE HUMIDITY: MIN / MAX _____ / _____ %				☐ OTHER _____			
UNUSUAL CONDITIONS: (6.1.29) ☐ DUST ☐ FUMES				REMARKS: _____			
☐ OTHER _____							

☐ UTILITY CONDITIONS: SEE GENERAL NOTES ON SHEET 6 *				☐ SYSTEM DESCRIPTION (6.2.2)			
STEAM: DRIVERS HEATING				SUCTION VESSEL: ☐ VENTED TO ATMOS ☐ CLOSED			
MIN _____ _____ _____				PUMP LOCATION: ☐ BELOW LIQ. LEVEL ☐ ABOVE LIQ. LEVEL			
MAX _____ _____ _____				SUCTION VESSEL ON LEVEL CONTROL? ☐ YES ☐ NO			
ELECTRICITY: DRIVERS HEATING CONTROL SHUTDOWN				PRESSURE SENSOR ON SUCTION VESSEL ☐ YES ☐ NO			
VOLTAGE _____ _____ _____				SUCTION VESSEL PRESSURE MAINTAINED BY: _____			
HERTZ _____ _____ _____							
PHASE _____ _____ _____							

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.



UOP LLC | 25 East Algonquin Road | Des Plaines, Illinois 60017-5017 USA

SEALLESS PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 506

SHEET 4 / 7

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	31-May-12	MW	MEH				
2	19-Jul-12	MW	DRL				
3	25-Sep-13	MW	CJH				

REV	☐ PERFORMANCE PROPOSAL CURVE NO. _____ ☐ RPM 2950 ☐ IMPELLER DIA.: RATED _____ MAX _____ MIN _____ mm ☐ HYD. EFFICIENCY AT RATED _____ 51 _____ % ☐ HYSTERESIS & MECH. LOSSES _____ 0.7 kW _____ kW ☐ OVERALL EFF. % 32.8 RATED POWER 2.3 kW ☐ MAX POWER @ RATED IMPELLER _____ kW ☐ MINIMUM CONTINUOUS FLOW: THERMAL _____ m³/h STABLE _____ m³/h ☐ ROTOR CHAMBER TEMP. RISE OPERATING (6.2.1) _____ °C ☐ ROTOR CHAMBER TEMP. RISE ON SHUTDOWN _____ °C ☐ PREFERRED OPER. REGION _____ TO _____ m³/h ☐ ALLOWABLE OPER. REGION _____ TO _____ m³/h LIMITED BY: ☐ VIBRATION ☐ OTHER _____ ☐ MAX HEAD @ RATED IMPELLER _____ m ☐ PERCENT RISE TO SHUTOFF (6.1.12) _____ ☐ ORIFICE USED TO STEEPEN CURVE OR GIVE CONT. RISE (6.1.12) ☐ NPSHR AT RATED CAPACITY (6.1.19) _____ m ☐ MAX SUCTION SPECIFIC SPEED ALLOW (6.1.10) 13,000 max (m³/hr, meters) ☐ SUCTION SPECIFIC SPEED QUOTED _____ ☐ MAX SOUND PRESS LEVEL REQ'D (6.1.15) _____ dBA ☐ EST. MAX SOUND PRESS LEVEL _____ dBA REMARKS _____	CONSTRUCTION - CONT'D ☐ NOZZLE CONDITIONS: (6.4.2) <table border="1" style="width:100%; border-collapse: collapse;"> <tr> <th>SIZE (NPS)</th><th>FLANGE CLASS</th><th>FAC'G</th><th>POSITION</th></tr> <tr> <td>SUCTION</td><td>150</td><td>RF</td><td></td></tr> <tr> <td>DISCHARGE</td><td>150</td><td>RF</td><td></td></tr> </table> ☐ PRESSURE CASING AUX. CONNECTIONS: (6.4.3) <table border="1" style="width:100%; border-collapse: collapse;"> <tr> <th>NO.</th><th>SIZE (NPS)</th><th>TYPE</th></tr> <tr> <td>☐ DRAIN</td><td></td><td>Provide with Valve</td></tr> <tr> <td>☐ VENT</td><td></td><td></td></tr> <tr> <td>☐ PRESSURE GAUGE</td><td></td><td></td></tr> <tr> <td>☐ TEMP GAUGE</td><td></td><td></td></tr> <tr> <td>☐ WARM-UP</td><td></td><td></td></tr> <tr> <td>☐ EXT. FLUSH/LEAK-OFF</td><td></td><td></td></tr> <tr> <td>☐ PURGE/FLUSHOUT</td><td></td><td></td></tr> </table> VENDOR SUPPLY VALVES FOR: (7.3.2.3) ☐ VENTS(S) ☐ DRAIN(S) ☐ ROTOR CAVITY DRAINABLE THRU SECONDARY DRAIN (6.1.8) ☐ CYLINDRICAL THREADS REQ'D FOR AUX. CONN'S (6.4.3.8.3) ☐ FLANGE THICKNESS REQ'S NON-STD BOLT LENGTH (6.4.2.3) ☐ THD CONN. TO SECONDARY CASING IS NOT ACCEPT. (6.4.3.8) SECONDARY CASING HAS: (6.8.8) ☐ DRAIN CONN. ☐ & FLUSH PROVISION	SIZE (NPS)	FLANGE CLASS	FAC'G	POSITION	SUCTION	150	RF		DISCHARGE	150	RF		NO.	SIZE (NPS)	TYPE	☐ DRAIN		Provide with Valve	☐ VENT			☐ PRESSURE GAUGE			☐ TEMP GAUGE			☐ WARM-UP			☐ EXT. FLUSH/LEAK-OFF			☐ PURGE/FLUSHOUT			1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
SIZE (NPS)	FLANGE CLASS	FAC'G	POSITION																																				
SUCTION	150	RF																																					
DISCHARGE	150	RF																																					
NO.	SIZE (NPS)	TYPE																																					
☐ DRAIN		Provide with Valve																																					
☐ VENT																																							
☐ PRESSURE GAUGE																																							
☐ TEMP GAUGE																																							
☐ WARM-UP																																							
☐ EXT. FLUSH/LEAK-OFF																																							
☐ PURGE/FLUSHOUT																																							
	CONSTRUCTION APPLICABLE STANDARD: ☐ API 685 ☒ OTHER ASME B73.3 (SEE REMARKS) ROTATION: (VIEWED FROM COUPLING END) ☐ CW ☐ CCW PUMP TYPE: ☒ OH1 ☐ OH2 ☐ OH5 ☐ OTHER _____ CASING MOUNTING: ☐ CENTERLINE ☐ NEAR CENTERLINE ☒ FOOT ☐ IN-LINE ☐ OTHER _____ CASING TYPE: ☒ SINGLE VOLUTE ☐ MULTIPLE VOLUTE ☐ DIFFUSER CASE PRESSURE RATING: ☐ MAX ALLOWABLE WORKING PRESSURE 14.06 kg/cm²g @ _____ 0 ☐ HYDROTEST PRESSURE 21.09 kg/cm²g DESIGN PUMP FOR: ☐ FULL ☐ _____ kg/cm²g VACUUM REMARKS _____	AUXILIARY PIPING (7.3) ☐ AUXILIARY PIPING PLAN: 23-S CIRC. PIPING MATERIALS: ☐ TUBING ☐ CARBON STEEL ☐ PIPE ☐ STAINLESS STEEL CIRC. PIPING ASSEMBLY: (7.5.2.10) ☐ THREADED ☐ UNIONS ☐ SOCKET WELDED ☐ FLANGED ☐ TUBE FITTINGS ☐ COOLING WATER PIPING PLAN (7.3.4) M-S C.W. PIPING ASSEMBLY: ☐ PIPE ☐ TUBING FITTINGS _____ C.W. PIPING MATERIALS: ☐ STAINLESS STEEL ☐ C. STEEL ☐ GALVANIZED COOLING WATER REQUIREMENTS: ☐ JACKET ☐ BEARING HOUSING _____ m³/h HEAT EXCHANGER _____ m³/h TOTAL COOLING WATER _____ m³/h STEAM PIPING: ☐ TUBING ☐ PIPE ☐ OTHER REMARKS _____	25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43																																				
	ROTOR: ☐ OPEN ☐ SEMI-OPEN ☐ FAB'D IMPELLERS ARE ACCEPT ☐ RENEWABLE CASE WEAR RINGS ARE NOT REQUIRED (6.7.1) ☐ RENEWABLE IMP. WEAR RINGS ARE NOT REQUIRED (6.7.1) ☐ NON-GROOVED BEARINGS ARE ACCEPTABLE (6.10.5) ☐ SHAFT DIAMETER BETWEEN BEARINGS _____ mm ☐ SPAN BETWEEN BEARING CENTERS _____ mm ☐ SPAN BETWEEN BEARING & IMPELLER _____ mm REMARKS _____	MATERIALS ☐ APPENDIX H CLASS (6.11.1) _____ ☐ MIN DESIGN METAL TEMP (6.11.4.5) 15 °C ☐ CASE Killed Carbon Steel IMPELLER Killed Carbon Steel ☐ CASE/IMPELLER WEAR RINGS _____ ☐ SHAFT 316 Stainless ☐ CONTAINMENT SHELL/STATOR LINER 316 Stainless ☐ INNER MAG. SHEATHING/ROTOR LINER _____ ☐ BEARINGS SiC ☐ OTHER _____	44 45 46 47 48 49 50 51 52 53																																				



UOP LLC | 25 East Algonquin Road | Des Plaines, Illinois 60017-5017 USA

SEALLESS PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 506

SHEET 5 / 7

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				

REV	SECONDARY CONTROL/CONTAINMENT ☐ SECONDARY CONTROL (6.8.1) ☐ MAX LEAKAGE ON PRIMARY FAILURE: _____ m³/h ☐ SECONDARY CONTAINMENT (6.8.1) AUX. SEAL DEVICE: ☐ MECHANICAL SEAL ☐ BUSHING ☐ OTHER _____ DEVICE IS: ☐ DRY RUNNING ☐ WET RUNNING ☐ INSIDE ☐ OUTSIDE OF BEARINGS (MAG-DRIVE) ☐ OTHER: _____ MECHANICAL SEAL DATA: ☐ APPENDIX H SEAL CODE _____ ☐ SEAL MANUFACTURER _____ ☐ SIZE AND TYPE _____ / _____ ☐ MANUFACTURER CODE _____ ☐ REMARKS _____	MAG-DRIVE SPECIFIC - CONT'D SHAFT COUPLINGS: (DRIVER-TO-PUMP) ☐ MAKE _____ ☐ MODEL Flexible with SS Disks ☐ NON-LUBE ☐ LUBRICATED: (SEE REMARKS) ☐ RATING (kW/100 RPM) _____ ☐ SERVICE FACTOR _____ ☐ LIMITED END FLOAT REQUIRED _____ ☐ SPACER, LENGTH _____ mm ☐ NOT REQUIRED ☐ MATERIAL: HUBS _____ SPACER _____ ELEMENTS _____ ☐ SHAFT DIAMETER AT COUPLING _____ mm DRIVER HALF COUPLING MOUNTED BY: ☐ PUMP MFR. ☐ DRIVER MFR. ☐ PURCHASER REMARKS _____ SEPARATE MOTOR DRIVE APPLICABLE SPEC: ☐ IEEE841 ☐ API541 ☐ OTHER: _____ ☐ FRAME _____ ☐ ENCLOSURE _____ ☐ LUBE _____ BEARING (TYPE/NUMBER): ☐ RADIAL _____ / _____ ☐ THRUST _____ / _____ REMARKS _____ BASEPLATES (9.1.5.3): ☐ API BASEPLATE NUMBER _____ (APPENDIX M) ☐ NON-GROUT CONSTRUCTION: (9.1.5.3.13) ☐ BASEPLATE MATERIAL _____ ☐ BOLT V-INLINE PUMP TO PAD/FOUNDATION (6.1.30.3) REMARKS _____	1																		
	MAG-DRIVE SPECIFIC ☐ DESIGN FOR REMOVAL OF DRIVE END WITHOUT DISTURBING THE PRESSURE CASING OR DRIVER (9.1.1.4) MAGNETIC COUPLING: ☐ SYNCHRONOUS ☐ TORQUE RING <table border="1" style="width:100%; border-collapse: collapse;"> <tr> <th style="width:50%;">MAGNETS:</th> <th style="width:50%;">OUTER</th> <th style="width:50%;">INNER</th> </tr> <tr><td>☐ MAG. MATERIAL</td><td></td><td></td></tr> <tr><td>☐ MOUNT. METHOD</td><td></td><td></td></tr> <tr><td>☐ TEMP. LIMIT. °C</td><td></td><td></td></tr> <tr><td>☐ HERMETIC SEALED</td><td></td><td></td></tr> <tr><td>☐ NO. OF MAGNETS</td><td></td><td></td></tr> </table> ☐ SOFT START REQ'D ☐ HARD START CAPABILITY ☐ TORQUE RATING (DECOUPLING), Nm _____ ☐ MAX TORQUE REQ'D ON STARTING, Nm (9.1.3.7a) _____ ☐ PUMP TORQUE AT RATED (+5%), Nm (9.1.3.7b) _____ ☐ ACTUAL SERVICE FACTOR (9.1.3.8) _____ % ☐ TORQUE REQ'D FOR FULL CURVE, Nm (9.1.3.7c) _____ ☐ DESIGN FOR FULL CURVE TORQUE REQUIREMENTS (9.1.3.7c) ☐ SUBMIT MAG-COUPLING TORQUE - TEMP CURVE (9.1.3.9) ☐ SUBMIT SPEED VS TORQUE CURVE (9.1.3.10) ☐ _____ EXTERNAL BEARING (TYPE/NUMBER): ☐ RADIAL Ball / _____ ☐ THRUST Ball / _____ LUBRICATION: (9.1.4.3) ☐ RING OIL ☒ FLINGER ☒ FLOOD ☐ PURGE OIL MIST ☒ PURE OIL MIST ☒ OTHER _____ ☐ CONSTANT LEVEL OILER PREFERENCE (SEE REMARKS) (9.1.4.2.1) ☒ OIL VISC. ISO GRADE _____	MAGNETS:	OUTER	INNER	☐ MAG. MATERIAL			☐ MOUNT. METHOD			☐ TEMP. LIMIT. °C			☐ HERMETIC SEALED			☐ NO. OF MAGNETS			CANNED MOTOR SPECIFIC ☐ SOLID MEDIA POTTING ALLOWED IN STATOR (9.2.2.2) ☐ CERAMIC INSULATION ACCEPTABLE (9.2.2.8) ☐ DESIGN MOTOR FOR FREQUENT STARTS (9.2.2.9) ☐ STARTS PER _____ ☐ IMPACT ON LIFE: (REMARKS) ☐ DESIGN MOTOR FOR: (9.2.2.9) ☐ TWO-SPEED ☐ VARIABLE SPEED _____ ☐ UL, FM OR EQUIVALENT REQ'D (9.2.2.10) ☐ UL ☐ FM ☐ OTHER ☐ CERTIFICATION OF IEEE 252 TEST REQUIRED (9.2.7) ☐ DECONTAMINATION CONNECTION ON STATOR (9.2.2.11) REMARKS _____	2
MAGNETS:	OUTER	INNER																			
☐ MAG. MATERIAL																					
☐ MOUNT. METHOD																					
☐ TEMP. LIMIT. °C																					
☐ HERMETIC SEALED																					
☐ NO. OF MAGNETS																					
			3																		
			4																		
			5																		
			6																		
			7																		
			8																		
			9																		
			10																		
			11																		
			12																		
			13																		
			14																		
			15																		
			16																		
			17																		
			18																		
			19																		
			20																		
			21																		
			22																		
			23																		
			24																		
			25																		
			26																		
			27																		
			28																		
			29																		
			30																		
			31																		
			32																		
			33																		
			34																		
			35																		
			36																		
			37																		
			38																		
			39																		
			40																		
			41																		
			42																		
			43																		
			44																		
			45																		
			46																		
			47																		
			48																		
			49																		
			50																		
			51																		
			52																		

ADDITIONAL REMARKS: Pumps shall comply with UOP Standard Specifications 7-12 and 5-16



UOP LLC | 25 East Algonquin Road | Des Plaines, Illinois 60017-5017 USA

SEALLESS PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 506

SHEET 6 / 7

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	08-May-12	MW	DRL				

REV

GENERAL NOTES:

- Vendor's offering shall be a unit of duplicate size and design which has a successful record of proven field service at operating conditions similar to those specified (proto-types are not acceptable). An installation list shall be submitted upon request. In the event no similar unit is available, the vendor may offer an alternate with a detailed explanation of where the offering differs from his field proven equipment.
- The design pressure of downstream equipment, piping, and instrumentation has been arrived at by adding 1.25 times the rated pump differential to the maximum suction pressure. Contractor to adjust as necessary or verify that downstream equipment, piping, and instrumentation design pressures are adequate for the shutoff pressure of purchased pumps.
- Flange classes are specified in accordance with ASME B16.1 and ASME B16.5. Flange classes listed in the UOP Project Specifications are based upon design pressure and temperature conditions only, and do not account for other loads. The final design of all flanges shall account for gasket seating and external loads. Differential thermal expansion for dissimilar joints and transient thermal conditions such as start-up/shutdown and operational upset shall be accommodated.
- All flanges intended for use with spiral wound gaskets shall have a flange surface finish of 125 Ra minimum to 250 Ra maximum. Finishes shall be judged by visual comparison with surface finish roughness standards conforming to ASME B46.1.
- Testing
 - a) Each pump shall have a factory performance test.
 - b) Pressure containing parts (including auxiliaries) shall be tested hydrostatically with liquid at a minimum of 1-½ times the maximum allowable working pressure.
- A self-cleaning type strainer shall be built into the pump discharge nozzle to prevent grit and other large size particles from passing through the flushing line and into the motor section.
- Alarms and Shutdowns
The following alarm and shutdown transmitters shall be provided:

	Alarms	Shutdown
Low discharge flow (by others)		
High/low motor power	X	X
High/low stator pressure (Canned type only)	X	
- Shutdown Circuitry shall be designed to stop the operating pump for an abnormal condition by any of the listed shutdown variables.
- Start circuitry shall be designed with a time delay to maintain control circuit energization until the discharge flow is established.
- Alarm circuitry shall be designed to series connect all listed alarm contacts, for each pump service, to activate a common trouble alarm (by others) in the control center.
- Contractor shall coordinate the details of the pump shutdown/start/alarm circuitry.

G1. Lubrication system is applied to high voltage motor's and all pumps' bearings except for vertical and diaphragm pumps.

* SEE BASIC ENGINEERING DESIGN QUESTIONNAIRE (BEDQ), ENGINEERING DESIGN INFORMATION (EDI) PROJECT 963763 A.4, SECTIONS 3 AND 4.

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.



UOP LLC | 25 East Algonquin Road | Des Plaines, Illinois 60017-5017 USA

SEALLESS PUMPS AND DRIVERS

PROJECT SPECIFICATION

963766 - 506

SHEET 7 / 7

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	09-Apr-12	MW	DRL				
1	16-Oct-13	MW	DRL				

REV

The following notes apply to those services for which they are specified in this Project Specification:

PUMP NOTES:

1. A spill back to suction source will be provided to insure satisfactory pump operation at minimum continuous flow. Pump vendor to advise minimum continuous capacity.
3. Normal, rated and other capacities include 2.27 m³/hr through restriction orifice-in spillback line.
5. The "capacity, other" is based on 50% of the lowest normal operating flow.

- C1. Variation in pumped fluid physical properties is listed below (Item No. P-2308A/B)

Temperature, °C	116.7	138.9	163.9
Vapor pressure, kg/cm ² [a]	1.88	3.573	6.954
Viscosity, cP	0.239	0.198	0.166
Specific Heat, kcal/(kg.°C)	1.012	1.024	1.041
Thermal Conductivity, kcal/(h.m.°C)	0.586	0.587	0.583

- C2. Variation in pumped fluid physical properties is listed below (Item No. P-2309A/B)

Temperature, °C	135.5	160.5	185.5
Vapor pressure, kg/cm ² [a]	3.28	6.394	11.601
Viscosity, cP	0.204	0.17	0.145
Specific Heat, kcal/(kg.°C)	1.021	1.038	1.061
Thermal Conductivity, kcal/(h.m.°C)	0.587	0.584	0.387

MOTOR NOTES:

- M10. Motor and coupling shall have the torque capability to accelerate the pump to operating speed with the discharge valve open, and the discharge check valve closed against the following static pressure:

Item Number	Minimum Autostart Pressure
P-2308A/B	2.25 kg/cm ² (g)
P-2309A/B	3.80 kg/cm ² (g)

- M12. Deleted.